

TrES-3b

R-2559

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110410 · created 2026-08-26T23:03:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455661.90012 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.057
Transit depth (Rp/Rs)^2	2.7 +/- 1.89 [%]
Orbital Inclination (inc)	82.46 +/- 2.89
Airmass coefficient 1 (a1)	0.913 +/- 0.012
Airmass coefficient 2 (a2)	0.08 +/- 0.27
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#1 - [50, 272]
Optimal Aperture	3.37
Optimal Annulus	6.62
Transit Duration (day)	0.051 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.164 $\pm$ 0.057 vs 0.16309 (deviation 0.02 σ)
Duration fit vs published	0.051 d vs 0.0567 d (deviation 0.21 σ)
Mid-time O-C	-2.9 min
Depth SNR	2.9
Residual scatter	1.29 %

Light curve

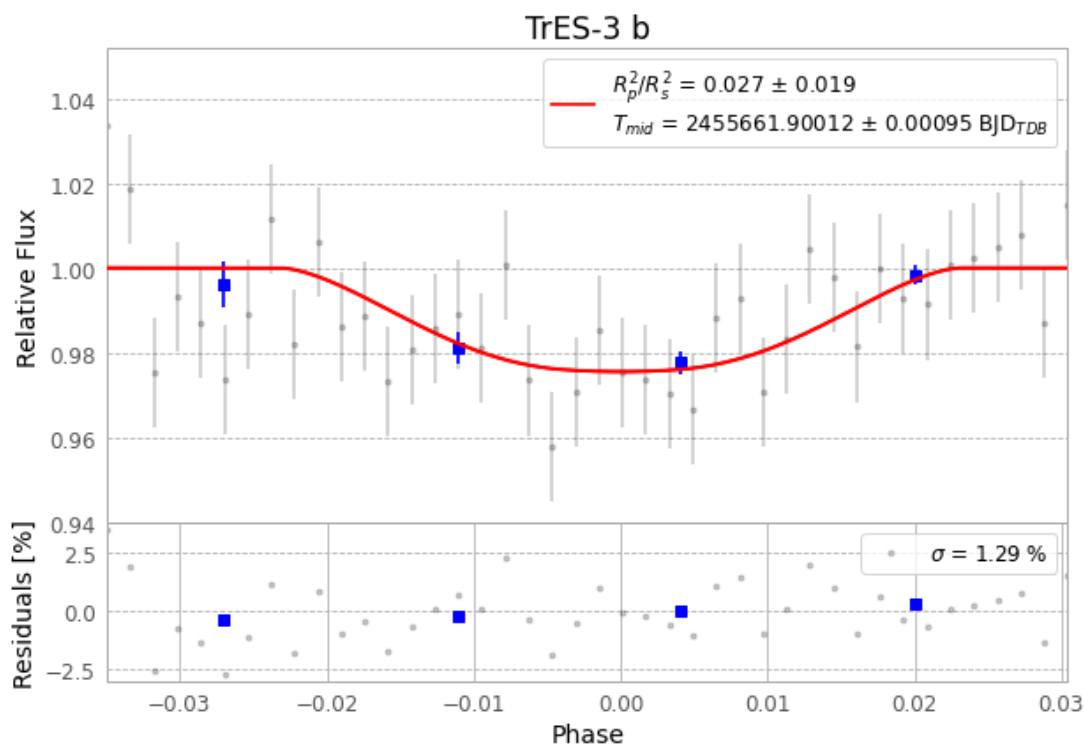


Figure 1 — FinalLightCurve_TrES-3 b_2011-04-10.png (SHA-256 a6ac63e1c7d5fc70...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 185], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f41ec7513685cb42...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), TRES-3110410080312.FITS → TRES-3110410103013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110410.log (a5a8309ae48d...), FinalLightCurve_TrES-3 b_2011-04-10.png (a6ac63e1c7d5...), FinalLightCurve_TrES-3 b_2011-04-10.csv (f77501cf0820...)

Compute resources

Wall time 369.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:03Z.